SUPERMARKET SALES ANALYSIS

FINAL DATA ANALYTICS PROJECT Internship Organization: SWYNEX

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Course: Bachelor of Computer Applications (BCA)

Project Tools: Microsoft Excel, Power BI

Project Type: Data Analytics and Visualization

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Supermarket Sales Analysis

1. Introduction

The Supermarket Sales Analysis project focuses on analyzing supermarket sales data to understand business performance, customer purchasing behaviour, product performance, and revenue trends.

The project uses data analytics techniques to clean, organize, analyze, and visualize sales data. Power BI is used to create an interactive dashboard that presents important business information through charts, graphs, and key performance indicators (KPIs).

This analysis helps identify sales patterns, understand customer preferences, compare branch performance, and support better business decision-making.

2. Problem Statement

Supermarkets generate large amounts of sales data every day. Without proper analysis, it becomes difficult to identify top-performing products, understand customer preferences, track revenue, and compare branch performance.

The objective of this project is to analyze supermarket sales data and develop an interactive dashboard that converts raw sales data into meaningful business insights.

3. Project Objectives

The main objectives of this project are:

1. To analyze supermarket sales data and understand overall business performance.
2. To identify the best-performing product lines.
3. To compare sales performance across different branches.
4. To analyze customer purchasing behaviour.
5. To understand popular payment methods.
6. To identify sales trends over different months.
7. To create an interactive Power BI dashboard.
8. To provide meaningful business insights for better decision-making.

4. Dataset Description

The dataset used in this project is the Supermarket Sales dataset. It contains sales transactions recorded across different supermarket branches.

The dataset includes information about products, customers, sales, payment methods, dates, and ratings.

Dataset Features:

- Invoice ID
- Branch
- City
- Customer Type
- Gender
- Product Line
- Unit Price
- Quantity
- Tax
- Total
- Date
- Time
- Payment
- COGS
- Gross Income
- Rating

Purpose of Dataset:

The dataset is used to analyze sales performance, customer behaviour, product demand, and branch-wise business performance.

5. Data Cleaning Process

Data cleaning is an important step in preparing the supermarket sales dataset for analysis.

The following steps were performed:

1. Checked the dataset for missing values.
2. Identified and removed duplicate records, if any.
3. Checked column names and data formats.
4. Converted the Date column into the correct date format.
5. Checked numerical columns for consistency.
6. Verified the dataset to ensure it was ready for analysis.

After data cleaning, the dataset was prepared for further analysis and visualization in Power BI.

6. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to understand the supermarket sales dataset and identify important patterns and trends.

The following analysis was carried out:

1. Analyzed total sales and revenue.
2. Compared sales across different branches.
3. Identified top-performing product lines.
4. Analyzed customer types and gender distribution.
5. Examined different payment methods.
6. Studied monthly sales trends.
7. Analyzed customer ratings and gross income.

EDA helped identify useful patterns and provided a foundation for creating the interactive Power BI dashboard.

7. Power BI Dashboard

An interactive dashboard was developed using Microsoft Power BI to visualize supermarket sales data.

The dashboard includes the following visualizations:

1. **Total Sales:** Displays the overall sales revenue.
2. **Total Orders:** Shows the total number of transactions.

3. **Quantity Sold:** Displays the total quantity of products sold.
4. **Average Rating:** Shows the average customer rating.
5. **Product Line Analysis:** Compares sales across different product categories.
6. **Branch-wise Sales:** Compares sales performance across branches.
7. **Payment Method Analysis:** Shows customer payment preferences.
8. **Monthly Sales Trend:** Displays changes in sales over time.

The dashboard helps users understand business performance through interactive charts, graphs, and key performance indicators (KPIs).

8. Key Business Insights

The supermarket sales analysis provides useful insights into business performance.

The following insights can be obtained from the dashboard:

1. Sales performance can be compared across different supermarket branches.
2. Product line analysis helps identify popular product categories.
3. Payment analysis shows which payment methods customers prefer.
4. Customer analysis helps understand purchasing behaviour.
5. Monthly sales trends help identify changes in business performance.
6. Customer ratings provide an understanding of customer satisfaction.
7. Gross income analysis helps evaluate business profitability.

These insights can help supermarket management improve sales strategies, understand customer needs, and make better business decisions.

9. Conclusion

The Supermarket Sales Analysis project provided practical experience in data cleaning, data analysis, and data visualization.

Using Microsoft Excel and Power BI, the sales dataset was analyzed and transformed into an interactive dashboard.

The project helped understand sales performance, customer purchasing behaviour, product demand, and branch-wise performance.

Overall, this project improved my data analytics skills and demonstrated how data visualization can support business decision-making.

10. Tools and Technologies Used

Microsoft Excel – Data cleaning and preparation.

Microsoft Power BI – Data visualization and dashboard creation.

Power Query – Data transformation.

GitHub – Project documentation and submission.